Causality - SoSe 2025

Graphical Models

Alexander Marx

Last Time

Variable Transformations

Theorem Let RV $X \sim p_X$ and RV $Z = t(X) \sim p_Z$ being defined via a function $t : \mathcal{X} \to \mathcal{Z}$. Then the PDFs are related as follows:

$$p_X(x) = p_Z(t(x))\ |t'(x)| = p_Z(t(x)) \left| \frac{dz(x)}{dx} \right|$$

$$p_Z(z) = p_X(t^{-1}(z))\ \left|(t^{-1})'(z)\right| = p_X(t^{-1}(z)) \left| \frac{dx(z)}{dz} \right|$$

where for the latter t must be invertible.

- For multivariate RVs, $\frac{dz(x)}{dx}$ refers to the **Jacobian matrix** and $|\cdot|$ is the **determinant** of the corresponding matrix.
- We can sample from distribution with CDF F by sampling from a uniform and **transforming** the variable with a **quantile function**.

Last Time

Sum and Product Rules

$$p(x) = \int p(x, y)\, dy \qquad \text{sum rule}$$

$$p(x, y) = p(x)\, p(y|x) \qquad \text{product rule}$$

Variations

$$p(y) = \int p(x, y)\, dx$$

$$p(x, y) = p(y)\, p(x|y)$$

$$p(x|y) = \frac{p(y|x)\, p(x)}{p(y)} \qquad \text{Bayes rule}$$

$$= \frac{p(y|x)\, p(x)}{\int p(y|x)\, p(x)\, dx} \qquad \text{combined}$$

Last Time

Definition (Independence) Two random variables X and Y are **statistically independent**, denoted by $X \perp\!\!\!\perp Y$ (or to be explicit $X \perp\!\!\!\perp Y$) if their joint PDF factorizes into the marginals, i.e.,

$$p(x, y) = p(x)p(y)$$

Definition (Expectation) The **expectation** of an expression involving random variables is the PDF-weighted integral:

$$\mathrm{E}[f(X)] = \int f(x)\, p(x)\, dx$$

Sometimes we omit the additional bracket: $\mathrm{E}\, f(X) = \mathrm{E}[f(X)]$

Outline

In today's lecture, we will connect graphs to probabilities. In particular, we discuss the following concepts:

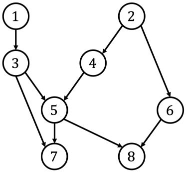
- Graph terminology
- Markov properties
- DAG models/Bayesian networks
- d-Separation

Graphs

Graph Terminology

Definition A **graph** $\mathcal{G} = (\boldsymbol{V}, \mathcal{E})$ consists of (finitely many):

- **Vertices** (nodes) $\boldsymbol{V} := \{1, \ldots, d\}$, and
- Directed **edges** $\mathcal{E} \subseteq \boldsymbol{V}^2$ with $(v, v) \notin \mathcal{E}$ for any $v \in \boldsymbol{V}$, e.g., $\mathcal{E} = \{(1,3), (2,6), \ldots\}$.

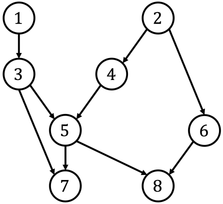


Further:

- Two nodes i and j are **adjacent**, if the edge $(i, j) \in \mathcal{E}$, or $(j, i) \in \mathcal{E}$.
- An edge between two adjacent nodes i and j is **undirected** if both $(i, j) \in \mathcal{E}$, **and** $(j, i) \in \mathcal{E}$.
- An edge between two adjacent nodes is **directed** if it is **not undirected**, e.g., if only $(i, j) \in \mathcal{E}$, and $(j, i) \notin \mathcal{E}$, then $i \to j$.
- If **all** edges are **directed**, the graph $\mathcal{G}$ is called **directed**.

Graph Terminology

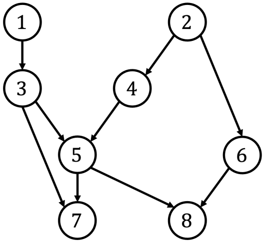
- A **path** between i and j is a sequence of **distinct** vertices $(i, \dots, j)$ such that **successive vertices are adjacent**.
- A **directed path** from i to j is a **path** between i and j where **all edges are pointing toward** j, i.e., $i \rightarrow \cdots \rightarrow j$.



- A **cycle** is a **path** $(i, j, \dots, k)$ **plus** edge between k and i.
- A **directed cycle** is a **directed path** $(i, j, \dots, k)$ with an additional edge $k \rightarrow i$.

Definition A **directed acyclic graph (DAG)** is a directed graph without directed cycles.

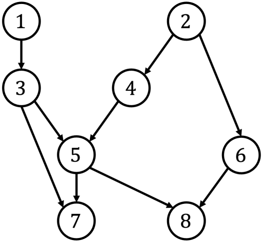
Examples



Which of the following are directed paths?

- (1,3,4,5)
  - *not a path since nodes 3 and 4 are not adjacent*
- (1,3,5,7,3,1)
  - *not a path since not all vertices are distinct*
- (5,7,3,1)
  - *a path but not directed*
- (2,6,8)
  - *a directed path*

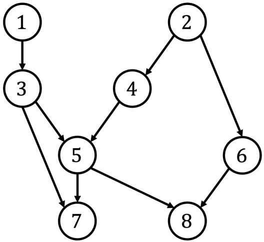
Examples



Cycles:

- $(3, 5, 7, 3)$ is a **cycle**, but *not a directed cycle*!
- $(2, 6, 8, 5, 4, 2)$ is a **cycle**, but *not a directed cycle*!
- This is a **DAG**.

Examples



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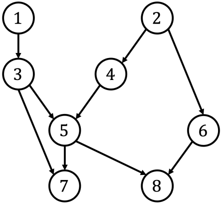
How can we modify this graph that it is *no longer* a DAG?

- E.g., we could swap the edge direction between 3 and 7 to get a directed cycle.

Graph Terminology

- If $i \to j$, then i is a **parent** of j, and j is a **child** of i.
- If there is a **directed path** from i to j, then i is an **ancestor** of j and j is a **descendant** of i.
- Each vertex is also an ancestor and descendant of itself (defined differently in the ECI book).



- Further, the **sets** of parents, children, descendants, and ancestors of i in $\mathcal{G}$ are denoted by $\mathrm{Pa}_{\mathcal{G}}(i), \mathrm{Ch}_{\mathcal{G}}(i), \mathrm{De}_{\mathcal{G}}(i), \mathrm{An}_{\mathcal{G}}(i)$.
- Whenever clear from context, we drop $\mathcal{G}$ in the notation, or abbreviate, e.g., $\mathrm{Pa}_{\mathcal{G}}(i)$ with Pa_i.

Graph Terminology

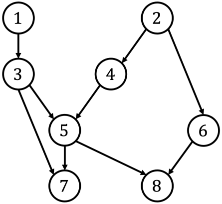
- We write sets of nodes/vertices in **bold face**.
- The previous definitions also apply for sets, e.g.

$$\mathrm{Pa}(\boldsymbol{S}) := \bigcup_{k \in \boldsymbol{S}} \mathrm{Pa}(k)\,.$$

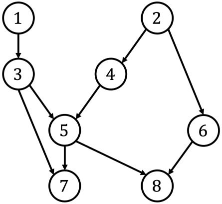
- The set of **non-descendants** of $\boldsymbol{S}$ is defined as the complement of the descendants, i.e.,

$$\mathrm{Nd}(\boldsymbol{S}) := \boldsymbol{V} \setminus \mathrm{De}(\boldsymbol{S})\,.$$

- We call a graph $\mathcal{G}$ **fully connected** if all pairs of nodes are adjacent.



Examples



Consider the graph on the left, what are:

- $\mathrm{Pa}(5)$?
  - $\{3, 4\}$
- $\mathrm{An}(5)$?
  - $\{1, 2, 3, 4, 5\}$
- $\mathrm{Ch}(4)$?
  - $\{5\}$
- $\mathrm{De}(4)$?
  - $\{4, 5, 7, 8\}$
- $\mathrm{De}(1, 2)$?
  - $\{1, 2, 3, 4, 5, 6, 7, 8\}$
- $\mathrm{Nd}(4)$?
  - $\{1, 2, 3, 6\}$
- $\mathrm{Nd}(3, 4)$?
  - $\{1, 2, 6\}$

Question

How many **fully connected DAGs** exist for a graph with d vertices?

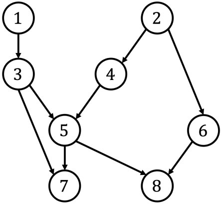
- For $d = 1$, there is only one possibility.
- For $d = 2$, we have $1 \rightarrow 2$, and $2 \rightarrow 1$ (2 options).
- For $d = 3$, there are 6 options, ...

In total, one can derive that it will be $d!$ possible DAGs!

Graphs and Random Variables

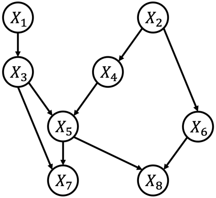
Graphs and Random Variables

- Each node **represents a random variable**, i.e., node i represents random variable X_i.
- If $\boldsymbol{A} \subseteq \boldsymbol{V}$, then $\boldsymbol{X_A} := \{X_i : i \in \boldsymbol{A}\}$.
- Edges denote relationships between pairs of variables (see following slides).



Note:

- When drawing graphs, we sometimes overload the notation and use the name of the RV as the name of the node, e.g., $X \rightarrow Y$, or as shown on the right.



Markovian Parents

As seen last time, we can always factorize the joint density of two variables X, Y with the product rule as

$$p(x, y) = p(x)p(y|x) = p(y)p(x|y) \; .$$

Let $\mathcal{G} = (\boldsymbol{V}, \mathcal{E})$ be a DAG with associated RVs $\boldsymbol{X}$, where the **nodes are ordered** s.t. $j > i$ cannot be an ancestor of i. We can factorize the joint density (via the **chain rule**) as

$$p(x_1, \ldots, x_d) = p(x_1)p(x_2|x_1) \cdots p(x_d|x_1, \ldots, x_{d-1}) \; .$$

Definition We call a set of variables $X_{\mathrm{Pa}(j)}$ the **Markovian parents** of X_j, if it is a **minimal** subset of $\{X_1, \ldots, X_{j-1}\}$ so that

$$p(x_j|x_1, \ldots, x_{j-1}) = p(x_j|x_{\mathrm{Pa}(j)}) \; .$$

Markov Factorization Property

Definition If a distribution P with density p factorizes as

$$p(x_1, \ldots, x_d) = \prod_{j=1}^{d} p(x_j | x_{\mathrm{Pa}(j)})$$

relative to DAG $\mathcal{G}$, then we say that P is **Markov** relative to $\mathcal{G}$.

If the equation above holds, then P is also said to fulfill the **Markov factorization property** with respect to $\mathcal{G}$.

Factorization of the Joint Density

Some Remarks:

- A distribution **can factorize according to several DAGs**, e.g., since

$$p(x_1, x_2) = p(x_1)p(x_2|x_1) = p(x_2)p(x_1|x_2) \ ,$$

  $p(x_1, x_2)$ factorizes according to the DAGs $1 \rightarrow 2$, **and** $1 \leftarrow 2$.
- Every distribution factorizes according to a **full DAG**.
  There are $p!$ possibilities!

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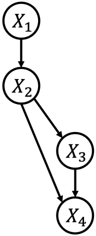
- Every distribution factorizes according to a **full DAG**.
  There are $p!$ possibilities!
- Sometimes, the **distribution factorizes w.r.t. a sparse DAG**, e.g.,

$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2|x_1) \cdot p(x_3|x_2)p(x_4|x_2, x_3)$$



DAG Models

Definition A **DAG model** or **Bayesian network** is a combination $(\mathcal{G}, P)$, where $\mathcal{G}$ is a DAG and P fulfills the Markov factorization property with respect to $\mathcal{G}$.

Bayesian networks can be used for various purposes:

- Estimating the joint density from low order conditional densities
- Reading off conditional independencies from the DAG
- Causal inference (cf. Pearl's causal hierarchy)

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Binary Variables

Imagine, you would like to estimate the density function for a set of d binary random variables. How many parameters do you need?

- **1 binary variable**: $p(X_1 = 1) = \pi$, and $p(X_1 = 0) = 1 - \pi$, i.e., **1 parameter**, which is π in this example
- **2 binary variables**: We need to express $p(1,1), p(1,0), p(0,1)$, $p(1,1)$, i.e., **3 parameters**
- **3 binary variables**: $p(1,0,1), \ldots$, i.e., **7 parameters**
- **d binary variables**: We need $2^d - 1$ **parameters**!

Assume that P **factorizes according to a chain graph**:
$1 \to 2 \to 3 \to \ldots$. How many parameters do we need?

- $2d - 1$ **parameters**! One parameter for $p(x_1)$ and two for each transition $p(x_i | x_{i-1})$.

Estimating Joint Densities

Whenever the distribution factorizes with respect to a sparse DAG, we only need to estimate

$$p(x_i | x_{\mathrm{Pa}(i)})$$

for each random variable i. Especially for **small parent sets**, this strongly reduces the number of parameters we need to estimate to fully describe $p(\boldsymbol{x})$.

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Conditional Independencies and the Markov Property

For **first-order Markov models**, the future is independent of the past given the present, i.e., $1 \to 2 \to \cdots \to (t-1) \to t \to (t+1)$, which implies that

$$X_{t+1} \perp\!\!\!\perp \{X_{t-1}, X_{t-2}, \ldots, X_1\} \mid X_t \ .$$

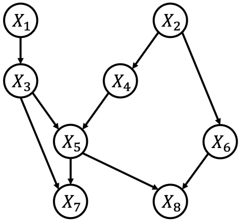
In DAGs, a similar property exists, called the **(local) Markov property**.

Definition (Local Markov Property) Given a DAG model $(\mathcal{G}, P)$ with $\mathcal{G} = (\boldsymbol{V}, \mathcal{E})$. Let $\boldsymbol{S} \subseteq \boldsymbol{V}$ then

$$\boldsymbol{X}_{\boldsymbol{S}} \perp\!\!\!\perp \boldsymbol{X}_{\mathrm{Nd}(\boldsymbol{S}) \setminus \mathrm{Pa}(\boldsymbol{S})} \mid \boldsymbol{X}_{\mathrm{Pa}(\boldsymbol{S})} \ .$$

In words: A set of variables $\boldsymbol{X}_{\boldsymbol{S}}$ is independent of all non-descendants (excluding their parents) given their parent set.

Example



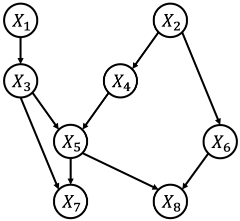
Let's define $\boldsymbol{S} = \{5\}$, and apply the local Markov property for this node:

- $\mathrm{Pa}(5) = \{3, 4\}$
- $\mathrm{Nd}(5) \setminus \mathrm{Pa}(5) = \{1, 2, 6\}$

Thus, for any distribution that factorizes according to the DAG on the left,

$$X_5 \perp\!\!\!\perp \{X_1, X_2, X_6\} \mid \{X_3, X_4\}$$

Example



Can we also determine if the following holds:

$$X_5 \perp\!\!\!\perp X_6 \mid X_8 \;?$$

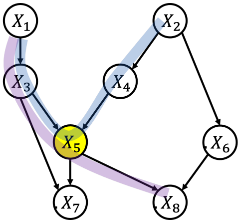
No, we cannot read off arbitrary conditional (in)dependencies based on the local Markov property ($6 \notin \mathrm{Pa}(5)$).

For this, we have **d-separation**!

Immortalities or Collider

Before we can define **d-separation**, we need one more definition.

> **Definition** A non-endpoint node k is a **collider on a path** p, if p contains the pattern "$\cdot \to k \leftarrow \cdot$". **Otherwise**, k is a **non-collider on the path** p.



Examples:

- 5 or (X_5) is a **collider** on the path $(1, 3, 5, 4, 2)$.
- 5 is a **non-collider** on the path $(1, 3, 5, 8)$.

d-Separation

Definition A **path** between i and j is **blocked by a set $\boldsymbol{S} \subseteq \boldsymbol{V}$** (not containing i or j) if

- there is a **non-collider** v on the path, and $v \in \boldsymbol{S}$, or
- there is a **collider** v on the path, and **neither** v **nor any of its descendants** are in $\boldsymbol{S}$.

A path that is **not blocked** is **active**.

Given two disjoint sets of nodes $\boldsymbol{A}, \boldsymbol{B} \subseteq \boldsymbol{V}$. If **all paths** between **any combination** of nodes $i \in \boldsymbol{A}$ and $j \in \boldsymbol{B}$ are **blocked by $\boldsymbol{S}$**, then **$\boldsymbol{A}$ and $\boldsymbol{B}$ are d-separated by $\boldsymbol{S}$**, written as $\boldsymbol{A} \perp\!\!\!\perp_{\mathcal{G}} \boldsymbol{B} \mid \boldsymbol{S}$.

Otherwise, $\boldsymbol{A}$ is **d-connected** to $\boldsymbol{B}$ given $\boldsymbol{S}$, denoted by $\boldsymbol{A} \not\!\perp\!\!\!\perp_{\mathcal{G}} \boldsymbol{B} \mid \boldsymbol{S}$.

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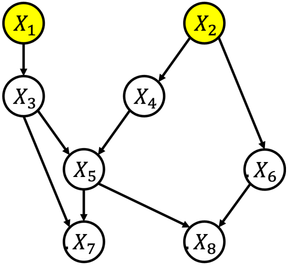
Note that $\perp\!\!\!\perp_{\mathcal{G}}$ is defined for **graphs** and should not be confused with $\perp\!\!\!\perp$, which relates to **statistical independence**.

Examples



Examples:

- $1 \perp\!\!\!\perp_{\mathcal{G}} 2$?
  - *True*
- $1 \perp\!\!\!\perp_{\mathcal{G}} 2 \mid 5$?
  - *False*
- $1 \perp\!\!\!\perp_{\mathcal{G}} 2 \mid 8$?
  - *False*
- $1 \perp\!\!\!\perp_{\mathcal{G}} 2 \mid \{3, 5\}$?
  - *True*

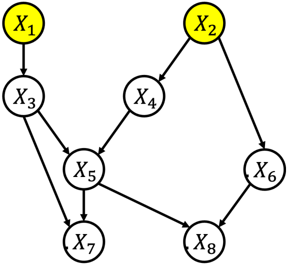
Global Markov Property

Definition Given a DAG $\mathcal{G} = (\boldsymbol{V}, \mathcal{E})$. A distribution P over $\boldsymbol{X}$ with density p satisfies the **global Markov property** with respect to $\mathcal{G}$ if for all disjoint subsets $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \subseteq \boldsymbol{V}$:

$$\boldsymbol{A} \perp\!\!\!\perp_{\mathcal{G}} \boldsymbol{B} \mid \boldsymbol{C} \implies \boldsymbol{X}_{\boldsymbol{A}} \perp\!\!\!\perp \boldsymbol{X}_{\boldsymbol{B}} \mid \boldsymbol{X}_{\boldsymbol{C}} \ .$$

Theorem (ECI book, 6.22) If $P_{\boldsymbol{X}}$ has density p, then the Markov factorization property (F), the local Markov property (L), and the global Markov property (G) are **equivalent**.

Examples



If $P_{\mathbf{X}}$ **satisfies the (global) Markov property** with respect to $\mathcal{G}$, then

- $X_1 \perp\!\!\!\perp X_2$
- $X_1 \not\perp\!\!\!\perp X_2 \mid X_5$
- $X_1 \not\perp\!\!\!\perp X_2 \mid X_8$
- $X_1 \perp\!\!\!\perp X_2 \mid \{X_3, X_5\}$

When we get to the lectures about **causal discovery**, we will need **assumptions about the inverse direction**. That is: *What does P tell us about $\mathcal{G}$?*

Other useful Rules for Proofs

Graphoid Axioms

Definition Let $\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z}$ be disjoint sets. The graphoid axioms are the following rules (where $\cdot \perp \cdot \mid \cdot$ is a ternary relation, e.g., $\perp\!\!\!\perp_{\mathcal{G}}$):

- **Symmetry** $\boldsymbol{X} \perp \boldsymbol{Y} \mid \boldsymbol{Z} \Rightarrow \boldsymbol{Y} \perp \boldsymbol{X} \mid \boldsymbol{Z}$.
- **Decomposition** $\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \mid \boldsymbol{Z} \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \mid \boldsymbol{Z}$.
- **Weak union** $\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \mid \boldsymbol{Z} \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \mid \boldsymbol{W} \cup \boldsymbol{Z}$.
- **Contraction**
  $(\boldsymbol{X} \perp \boldsymbol{Y} \mid \boldsymbol{W} \cup \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \mid \boldsymbol{Z}) \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \mid \boldsymbol{Z}$.
- **Intersection**
  $(\boldsymbol{X} \perp \boldsymbol{Y} \mid \boldsymbol{W} \cup \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \mid \boldsymbol{Y} \cup \boldsymbol{Z}) \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \mid \boldsymbol{Z}$, for any pairwise disjoint subsets $\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z} \subseteq \boldsymbol{V}$.

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One can show (see Pearl book, Chapter 1) that all **graphoid axioms hold for** $\perp\!\!\!\perp_{\mathcal{G}}$, whereas, the **intersection** property is **not fulfilled by arbitrary distributions** P. Thus, generally, only the first four properties hold for $\perp\!\!\!\perp$.

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Bayesian networks can be used for various purposes:

- Estimating the joint density from low order conditional densities
- Reading off conditional independencies from the DAG
- **Causal inference (cf. Pearl's causal hierarchy)**

Next time!

Credits

- **Recommended reading** is *Elements of Causal Inference* Chapter 6.1-2, and 6.5, or Chapter 1.1-2 in *Causality* (Pearl, 2009)
- This course *builds upon and extends* the material from *"Causality" (WiSe 2022/23)* that was kindly provided by **Stefan Harmeling**.
- This lecture, *builds upon and extends* the slides of the course "*Causality*" (2021) by Christina Heinze-Deml.

Thank you!